NATIONAL OPEN UNIVERSITY OF NIGERIA

PLOT 91, CADASTRAL ZONE, NNAMDI AZIKIWE EXPRESSSWAY, JABI-ABUJA

FACULTY OF SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

2023_2 EXAMINATIONS

COURSE CODE: CIT308

COURSE TITLE: FORMAL METHODS AND SOFTWARE DEVELOPMENT

COURSE CREDIT: 3 UNITS

TIME ALLOWED: 2 HOURS

INSTRUCTION: ANSWER QUESTION 1 AND ANY OTHER THREE QUESTIONS

Q1

(a) Highlight 8 requirements of user interface

4 marks

(b) Draw a level 0 data flow diagram for hotel reservation system?

5 marks

(c) Construct a truth table for the formula $((a \rightarrow b) \wedge (b \rightarrow c)) \leftrightarrow \neg a$

12 marks

(d) Enumerate 8 advantages of using formal methods?

4 marks

Q2

(a) Show that the inverse and the converse of a conditional are logically equivalent

8 marks

(b) Explain Universal and Existential statements?

7marks

Q3

(a) Complete the table comparing the three testing models under the highlighted areas?

Areas	Unit Testing	System Testing	Integration Testing
Objective			
Scope			
Test Environment			
Dependencies			

12 marks

(b) Explain model checking as a formal methods technique?

3 marks

Q4

(a) Complete the table to compare V-model with Spral model using any 4 differences

Differences	V-Model	Spiral Model
Development Approach		
Emphasis		
Testing Integration		
Feedback from Customer		
Change Management		
Project Size and Complexity		
Risk Management		
Development Flexibility		
Planning and Documentation		
Project Cost and Complexity		

12 marks

(b) Using a diagram only, describe the big-bang model?

3 marks

Q5

(a) Given: X={4, 7, 3, 2, 6}; Y={3, 2, 7, 2, 1} and X={ 8, 10}. Describe and compare $(X \cap Y) \cap Z$ and $(X \cup Y) \cup Z$

12 marks

(b) Using a diagram, describe forward engineering?

3 marks

Q6

(a) Complete any 6 dissimilarities in the table for universal and existential statements below:

Differences	Universal Statements	Existential Statements
Quantifier		
Assertion		
Scope		
Existential Import		

Differences	Universal Statements	Existential Statements
Negation		
Generalization		
Symbolic Representation		

12 marks

(b) List 6 contents of a test plan?

3 marks